

Zihao Zhu

☎ (86)166-1982-1803 • ✉ zihaozhu@link.cuhk.edu.cn • 🌐 zihao-ai.github.io
📄 Google Scholar (800+ Citations)

Education

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

Ph.D. in Data Science, School of Data Science

2021–2026 (Expected)

Advisor: Baoyuan Wu

University of Chinese Academy of Sciences

Beijing, China

M.S. in Cybersecurity, Institute of Information Engineering

2018–2021

Advisor: Jing Yu

China University of Mining and Technology

Xuzhou, China

B.Eng. in Computer Science and Technology, School of Computer Science and Technology 2014–2018

Research Interests

Large Language Model Safety, Reasoning Models, Embodied AI Agents, Data Governance, AI Security & Privacy, Multimodal Learning

Research Experience

Huawei 2012 Lab (University-Industry Collaboration)

Shenzhen, China

Research Collaboration

Dec. 2024–Present

- Investigated the “unthinking vulnerability” in reasoning models. Proposed the Breaking of Thought (BoT) attack and designed the Monitoring of Thought (MoT) framework to dynamically terminate redundant or risky reasoning.
- Proposed Adversarial Chain-of-Thought Tuning (AdvChain) to address the “snowball effect” in safety alignment of reasoning models, improving robustness against jailbreak attacks and CoT hijacking.

Tencent AI Lab

Shenzhen, China

Research Intern

Mar. 2024–July 2024

- Led the design of EARBench, the first physical risk awareness evaluation benchmark for embodied AI. Built EARDataset using multi-agent collaboration to generate risk scenarios for task planning safety evaluation.

Publications

Published 14 journal/conference papers (6 as first/co-first author); 10 preprints (6 as first author)

(TPAMI: 2; IJCV: 2; NeurIPS: 3; ICLR: 1; IJCAI: 2; AAAI: 1; ICASSP: 2; PR: 1)

Large Language Model Safety & Chain-of-Thought

- To Think or Not to Think: Exploring the Unthinking Vulnerability in Large Reasoning Models
Zihao Zhu, Hongbao Zhang, Ruotong Wang, Xu Ke, Siwei Lyu, Baoyuan Wu (NeurIPS 2025 Workshop)
- Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers
Huang et al., **Zihao Zhu**, Bernard Ghanem, Philip Torr, Guohao Li (NeurIPS 2025 Workshop)

3. AdvChain: Adversarial Chain-of-Thought Tuning for Robust Safety Alignment of Large Reasoning Models
Zihao Zhu, Xinyu Wu, Gehan Hu, Siwei Lyu, Ke Xu, Baoyuan Wu (ICLR 2026)

AI Agents

4. EARBench: Towards Evaluating Physical Risk Awareness for Task Planning of Foundation Model-based Embodied AI Agents
Zihao Zhu, Bingzhe Wu, Zhengyou Zhang, Lei Han, Qingshan Liu, Baoyuan Wu
5. BrandFusion: A Multi-Agent Framework for Seamless Brand Integration in Text-to-Video Generation
Zihao Zhu, Ruotong Wang, Siwei Lyu, Baoyuan Wu (CVPR 2026 Findings)

Data Governance

6. VDC: Versatile Data Cleanser based on Visual-Linguistic Inconsistency by Multimodal Large Language Models
Zihao Zhu, Mindda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu (ICLR 2024)
7. HID: A Hierarchical Framework for Multi-Granularity Visual-Textual Inconsistency Detection
Zihao Zhu, Hongbao Zhang, Guanzong Wu, Siwei Lyu, Baoyuan Wu

AI Security

8. Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective
Baoyuan Wu*, **Zihao Zhu***, Li Liu, Yanbo Fan, Siwei Lyu (*Co-first author with advisor) (IJCV 2025)
9. Defenses in Adversarial Machine Learning: A Survey
Baoyuan Wu, Mingli Zhu, Meixi Zheng, **Zihao Zhu**, Shaokui Wei, Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu (IEEE TPAMI 2025)
10. BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks
Meixi Zheng, Xuanchen Yan, **Zihao Zhu**, Hongrui Chen, Baoyuan Wu (IEEE TPAMI 2025)
11. BackdoorBench: A Comprehensive Benchmark and Analysis of Backdoor Learning
Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Mingli Zhu, Ruotong Wang, Li Liu, Chao Shen (IJCV 2025; NeurIPS 2025, 217 citations)
12. Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers
Ruotong Wang, Hongrui Chen, **Zihao Zhu**, Li Liu, Baoyuan Wu (Submitted to IEEE TPAMI)
13. Boosting Backdoor Attack with A Learnable Poisoning Sample Selection Strategy
Zihao Zhu, Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu (Submitted to Neurocomputing)

Multimodal Learning

14. From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering
Zihao Zhu (ICASSP 2022)
15. Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering
Zihao Zhu, Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu (IJCAI 2020, 177 citations)
16. Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering
Jing Yu*, **Zihao Zhu***, Yujing Wang, Weifeng Zhang, Yue Hu (*Co-first author with advisor) (Pattern Recognition 2020, 132 citations)

Other Publications

17. Learning to Optimize Permutation Flow Shop Scheduling via Graph-based Imitation Learning
*Longkang Li, Siyuan Liang, **Zihao Zhu**, Xiaochun Cao, Chris Ding, Hongyuan Zha, Baoyuan Wu (AAAI 2024)*
18. MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension
*Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, **Zihao Zhu**, Yajing Sun (ICASSP 2021)*
19. DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses
*Xiaoze Jiang, Jing Yu, Zengchang Qin, **Zihao Zhu**, Qi Wu (IJCAI 2020)*

Academic Service

Reviewer: IEEE TIP, ICLR, ICML, NeurIPS, CVPR, ICCV, AAAI, AISTATS

Awards & Honors

2024: Guotai Junan Scholarship

2021: AIRS Talent of PhD Research Program

2020: University of Chinese Academy of Sciences Merit Student

2018: Outstanding Undergraduate Thesis Award

2015, 2016: The First Prize Scholarship of CUMT